Closing the Variance Gap: Spectral PINN for Lévy-Driven SPDEs

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Abstract

Stochastic partial differential equations (SPDEs) driven by Lévy processes are central to mathematical finance, biology, and statistical physics, yet computing their solutions with quantified uncertainty remains expensive. Physics-informed neural networks (PINNs) offer a mesh-free alternative for deterministic PDEs, but on stochastic systems vanilla PINN training collapses to low-variance solutions: the predicted variance is underestimated by a factor of 4–5×, making the method scientifically unsuitable for uncertainty-driven applications. A natural counter-proposal — replacing the deterministic network with a Bayesian one, so that posterior weight uncertainty generates the missing variability — also fails: a mean-field Bayes-by-Backprop Bayesian PINN (the most widely used variational variant) is 7–33× worse than the deterministic vanilla baseline on log-spatial PSD MSE across all three of our benchmarks. More expressive Bayesian methods (HMC, deep ensembles) may behave differently and we treat their evaluation as future work. We introduce *Spectral PINN*, a drop-in modification that augments the standard PINN loss with three distributional regularisers — log-spatial PSD, log-temporal PSD, and variance ratio — evaluated on a fixed sub-grid against finite-difference reference solutions. The construction is theoretically

motivated by Bochner's theorem, which guarantees that for stationary Gaussian random fields with separable space-time covariance, matching spectral marginals recovers the full second-order law; for Lévy-driven fields it controls the second-order structure exactly and leaves higher cumulants as future work. On a hierarchical 3-channel SPDE driven by Variance-Gamma jumps (64×64 spatial grid, 512 timesteps), Spectral PINN reduces spatial and temporal log-PSD MSE by 3–4$\times$ over the vanilla PINN across all channels and closes the variance ratio gap from 0.19 to 0.83 (target: 1.0). On Burgers-VG the variance ratio rises from 0.50 to 0.93. On Heston-VG it closes the gap on the price channel and saturates at 0.14 on the variance channel; we document this structural failure mode in the discussion. The contribution is a simple, theoretically motivated recipe for uncertainty-aware neural learning of jump-diffusion SPDEs, with direct applications in derivative pricing and risk management.
Keywords: physics-informed neural networks, stochastic partial differential equations, Lévy processes, spectral methods, Bayesian neural networks

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1. Introduction

1.1. Motivation

Partial differential equations driven by Lévy processes have emerged as a vital mathematical framework across diverse disciplines, including financial risk management, statistical mechanics, and plasma physics — specifically in modelling non-local heat transport within Tokamak reactors. In contrast to Gaussian-based models, Lévy-driven dynamics inherently capture jump discontinuities and heavy-tailed distributions, which are essential for describing

extreme climatic events and biological foraging strategies under the Lévy Flight Foraging Hypothesis.

The numerical treatment of such non-local operators remains a significant challenge. Traditional approaches such as the Finite Difference (FD) method and Monte Carlo (MC) simulations serve as the current gold standard for accuracy and are employed in this study for benchmarking. Both, however, suffer from prohibitive computational costs: FD schemes lead to dense matrix representations due to non-locality, while MC methods exhibit $O(n^{-1/2})$ convergence, rendering them impractical for real-time applications where decision latency is critical.

1.2. The Problem

Deterministic PDEs make great use of the classical Laplacian operator, which depends only on a local neighbourhood. SPDEs of the form considered here are defined with the fractional Laplacian, which in contrast to its classical counterpart is an integral operator, requiring access to all field values in the domain. A direct mesh-based discretisation produces dense matrices and erodes the efficiency of grid methods.

The issue worsens with the introduction of a PINN. The behaviour of a neural network during training can be described using the Neural Tangent Kernel (NTK), whose eigenvalue spectrum determines the speed at which the network learns different frequency components of the target. For standard training, low-frequency eigenvalues dominate; the network learns the "general" shape of the solution and omits the high frequencies that encode Lévy jumps. The non-local fractional Laplacian compounds this problem, since the PINN tries to impose a non-local operator onto a local, smooth

neural-net structure.

1.3. Contribution

We propose the *Spectral PINN*, a drop-in modification to the standard PINN that augments the conventional loss with three distributional regularisers evaluated on a fixed sub-grid against a finite-difference reference: spatial power spectral density, temporal power spectral density, and variance ratio. The construction is theoretically motivated by Bochner's theorem, which guarantees that for stationary Gaussian random fields with separable space-time covariance, matching spectral marginals recovers the second-order law.

The contributions of this paper are:

- **Identification of the failure mode.** We quantify vanilla PINN variance underestimation on Lévy SPDEs across three problem classes: a factor of $\sim 5\times$ on the hierarchical 3-channel Fokker-Planck system, $\sim 2\times$ on Burgers-VG, and a complete variance collapse (vr ≈ 0) on the Heston-VG price channel.

- **Spectral PINN method.** A simple drop-in modification: three distributional regularisers grounded in Bochner's theorem, with no change to architecture, optimiser, or training data.

- **Bayesian baseline.** A mean-field Bayes-by-Backprop PINN trained at identical architectural parameters fails by 7–33$\times$ on log-spatial PSD MSE relative to the deterministic vanilla baseline, demonstrating that variational uncertainty quantification over network weights does not substitute for an explicit distributional loss.

- **Theoretical scope.** A structural argument (built on Bochner's theorem and Whittle log-likelihood asymptotics) showing the loss targets the correct quantities for stationary Gaussian fields with separable space-time covariance, and controls the second-order structure of stationary Lévy fields. A quantitative Plancherel–Wasserstein bound and a Mishra–Molinaro-style consistency theorem are deferred to future work.

- **Empirical validation.** Forty trained checkpoints across three Lévy-SPDE classes, demonstrating consistent variance-gap closure: $0.19 \to 0.83$ on the most-driven hierarchical channel, $0.50 \to 0.93$ on Burgers-VG, and $0.00 \to 0.48$ on the Heston-VG price channel.

- **Per-component ablation.** A 5-variant $\times$ 5-seed ablation on Burgers-VG demonstrates that the three loss terms are nearly orthogonal — each targets its own metric, single-term variants are unstable, and only the full combination achieves uniform improvement (Section 5.3).

- **Operator surrogate extension.** The same loss applied to a Fourier Neural Operator (companion paper) closes the variance gap on four of five test problems, with the largest gain a sixteen-fold closure on Heston-VG.

- **Honest characterisation.** An explicit account of where Spectral PINN provides marginal benefit (Heston variance channel, y-channel of the hierarchical SPDE) and the structural reason in each case.

1.4. Gap analysis

The combination of Lévy-driven SPDE and single-realisation neural surrogate sits at the intersection of four established research threads. None addresses the variance gap directly, and we show below — both by argument and, in the case of Bayesian PINNs, by direct experiment — that none can be retrofitted to do so without a structural change to the loss.

Standard physics-informed networks.. The original PINN formulation [1] minimises a sum of mean-squared collocation, initial, boundary, and data residuals. On deterministic PDEs the formulation is competitive with classical solvers; on stochastic problems it inherits the spectral-bias pathology of MLPs [2, 3]. Empirically, a vanilla PINN trained on our 2D hierarchical SPDE recovers a pointwise relative L^2 error of ≈ 0.3 on the most-driven channel but reaches only $\mathrm{Var}(u_\theta)/\mathrm{Var}(u_{\mathrm{FD}}) \approx 0.19$ — it averages across the noise realisation rather than reproducing it. Architectural fixes such as random Fourier features [3], adaptive sampling [4], NTK reweighting [5], or causal training [6] reduce pointwise error but leave the variance gap essentially untouched, because none of these losses penalises the spectrum of the predictor.

Bayesian PINNs.. The natural counter-proposal is that the variance gap is a symptom of unmodelled uncertainty: replace the deterministic network with a probabilistic one and let the posterior over weights generate the missing variability. Yang, Meng and Karniadakis [7] introduced B-PINNs along these lines, using Hamiltonian Monte Carlo or mean-field variational inference. We test the variational version directly: on both Burgers-VG and the

2D hierarchical SPDE, a mean-field Bayes-by-Backprop B-PINN with the *same* architecture as the deterministic baseline produces a posterior mean that is 7–33$\times$ worse than the deterministic vanilla PINN on log-spatial PSD MSE (Section 5.5, Figure 3). The variance ratio of the posterior mean collapses to ~ 0 on driven channels: the predictor is essentially the trivial zero function. The posterior standard deviation is non-trivial ($\sim$ 0.11–0.31) but reflects parameter uncertainty rather than the aleatoric noise structure of the SPDE driver. The mechanism is straightforward — the KL prior on weights regularises the posterior mean toward smoothness, removing exactly the high-frequency content needed to match the SPDE spectrum. Mean-field variational uncertainty quantification, by itself, cannot close the variance gap; whether more expressive posteriors (HMC, deep ensembles) behave differently is an open empirical question we leave to future work.

Neural operators and neural spectral methods.. A second family replaces the per-realisation network with an operator-learning surrogate trained on many input–output pairs: FNO [8], DeepONet [4], and the neural-spectral-methods family of Du et al. [9] which parameterises the network output directly in a Fourier basis. These learn the solution map $u_0 \mapsto u(\cdot, T)$ amortised across initial conditions or forcing realisations, and inherit the spectrum of their training distribution. Two issues prevent direct application here. First, training requires hundreds to thousands of full FD solutions — orders of magnitude more compute than the single realisation we use. Second, the output spectrum is tied to the empirical distribution of training samples, not to the law of any particular SPDE; extending to a new noise statistic (e.g. switching VG parameters) requires retraining. The Lévy-PINN results of

Zhang et al. [10] achieve higher variance ratios on VG-driven problems than our vanilla baseline by combining a moment-matching loss with a tailored architecture; we view their contribution as complementary to ours, addressing the same pathology along an architectural axis where ours is loss-only.

Spectral and distributional losses elsewhere.. Spectral losses appear in image generation [11], audio modelling [12], and turbulence closure [13], typically as auxiliary terms to a primary reconstruction objective. None has been integrated with a PDE residual on a stochastic problem; the closest is the Whittle-likelihood approach in classical time-series [14], which provides the asymptotic justification for our log-periodogram penalty (Section 3) but has not previously been combined with collocation losses.

Concurrent and architectural approaches.. Two recent works target the same failure mode along different axes. Worrall et al. [15] introduce "spectral shaping", an architectural reparameterisation that biases the network to allocate capacity to high-frequency modes; their approach is complementary to ours, modifying the network rather than the loss, and the two could be combined. Hu et al. [16] solve Fokker–Planck–Lévy systems with a score-based variant of the PINN that directly parameterises $\nabla \log p$; the spectral gap they encounter is structurally different from ours because the score-network output is a derivative of the density. We view both as parallel rather than competing contributions.

Synthesis.. The variance gap is not a side-effect of inadequate architecture, inadequate training, or inadequate uncertainty quantification — it is a consequence of training on losses that do not constrain the spectrum of the predic-

tor. Once the loss is augmented with explicit log-PSD and variance penalties, the gap closes ($0.50 \to 0.93$ on Burgers-VG; $0.19 \to 0.83$ on the most-driven 2D channel) using a network architecturally identical to the vanilla baseline. The contribution of this paper is not a new architecture, a new optimiser, or a new probabilistic framework, but the demonstration that the right loss — distributional, spectral, single-realisation — is sufficient.

1.5. Section roadmap

The remainder of the paper is organised as follows. **Section 2** introduces the Spectral PINN: the variance-gap diagnostic, the three distributional loss components, the total objective, and the Fourier-feature MLP architecture. **Section 3** develops the analytical motivation: Bochner's representation of stationary fields under the separability assumption used throughout, the Whittle-likelihood justification for the log scale (including its irreducible asymptotic bias floor), a structural argument that the log-PSD loss targets the correct second-order quantities, and the explicit gap for higher cumulants on Lévy-driven fields. **Section 4** specifies the three benchmark problems, the finite-difference reference solver, the four metrics used throughout (variance ratio, log-spatial / log-temporal PSD MSE, relative L^2), and the baselines (vanilla PINN, B-PINN). **Section 5** reports results, headlined by a $113\times$ gap between Spectral PINN and B-PINN on the most-driven channel of the 2D hierarchical SPDE. **Section 6** discusses limitations — most notably the Heston-VG variance channel and the y-channel seed outlier — and **Section 7** concludes.

Spectral PINN pipeline

u_θ (trainable) $\xrightarrow{\text{sub-grid}} \widehat{S}_{xy}, \widehat{S}_t, v_\theta$

u_{FD} (fixed) $\xrightarrow{\text{sub-grid}} S^*_{xy}, S^*_t, v^*$

$\mathcal{L} = \mathcal{L}_{\mathrm{xy}} + \mathcal{L}_t + \mathcal{L}_{\mathrm{var}}$ (Bochner-matched)

Figure 1: Spectral PINN: at each epoch the network u_θ is evaluated on a fixed sub-grid $\mathcal{G}_s$ of the FD reference grid. Spatial PSD, temporal PSD, and variance are computed by FFT and compared to precomputed targets in log space. The total loss augments the standard PINN objective with the resulting three distributional terms (eq. 3).

2. Method

2.1. What goes wrong with the standard PINN

Train a vanilla PINN on a Lévy-driven SPDE and inspect three diagnostics on a held-out grid:

1. Pointwise relative L^2 error: ≈ 0.3 — looks fine.
2. Variance ratio $\operatorname{Var} u_\theta / \operatorname{Var} u_{\mathrm{FD}}$: ≈ 0.2–0.5.
3. Power spectrum $|\mathcal{F} u_\theta|^2$ vs $|\mathcal{F} u_{\mathrm{FD}}|^2$: high modes are missing by 2–4 decades.

The network is averaging across the noise realisation rather than reproducing it. Pointwise loss is insensitive to high-frequency content because high-frequency error and high-frequency signal cancel almost exactly under MSE. Any sample-path-style use of u_θ — extracting tail statistics, computing exit times, valuing a path-dependent option — fails. We close this gap by adding three loss terms that directly penalise the missing structure.

2.2. Three terms, three diagnostics

Compute the FD reference once and extract three statistics on a sub-grid $\mathcal{G}_s$ of resolution $24 \times 24 \times 96$: the spatial power spectrum $S^*_{xy}(k)$, the temporal power spectrum $S^*_t(k)$, and the total variance v^*. These are detached constants from the optimiser's perspective. At every epoch we evaluate u_θ on $\mathcal{G}_s$ and form the matching statistics S_{xy}, S_t, v_θ. The new loss components are

$$\mathcal{L}_{\mathrm{xy}} = \overline{\left(\log_{10} S_{xy} - \log_{10} S^*_{xy}\right)^2}, \qquad \mathcal{L}_{\mathrm{t}} = \overline{\left(\log_{10} S_t - \log_{10} S^*_t\right)^2},$$
$$\mathcal{L}_{\mathrm{var}} = \left((v_\theta - v^*)/v^*\right)^2. \tag{1}$$

$\mathcal{L}_{\mathrm{xy}}$ and $\mathcal{L}_{\mathrm{t}}$ correct *shape*; $\mathcal{L}_{\mathrm{var}}$ corrects *scale*. The spatial PSD is computed symmetrically along both spatial axes,

$$\widehat{S}_{xy}(f) = \tfrac{1}{2}\Big(\overline{|\mathcal{F}_x f|^2}_{y,t} + \overline{|\mathcal{F}_y f|^2}_{x,t}\Big), \tag{2}$$

so anisotropic features that align with one axis do not bias the metric. The temporal PSD $\widehat{S}_t(f) = \overline{|\mathcal{F}_t f|^2}_{x,y}$ is averaged over space.

2.3. Why log, why MSE, why a sub-grid

Log scale. Power spectra span 4–6 orders of magnitude on Lévy problems. A linear loss would care only about the lowest few wavenumbers — exactly the regime the vanilla PINN already gets right. Working in $\log_{10}$ assigns equal weight to each decade and makes the high-mode discrepancy visible to the optimiser.

MSE in log space. Equivalent to a symmetric Bregman divergence between the two empirical spectral measures up to an additive constant, and equivalent to a Whittle-type negative log-likelihood for the periodogram [14,

17]. The minimiser is u_{FD} itself: the loss is consistent in the sense that no other matching measure achieves zero (modulo the asymptotic bias floor discussed in Section 3).

Sub-grid. The FFTs cost $O(n_x n_y n_t(\log n_x + \log n_y + \log n_t))$ per epoch. Using the full FD grid would dominate the wall time; the $24 \times 24 \times 96$ sub-grid keeps spectral evaluation under 5 % of total compute while preserving all wavenumbers up to the Nyquist frequency of $\mathcal{G}_s$, which empirically captures the regime where the variance gap is generated.

2.4. Why this is the right object

Three observations make $\mathcal{L}_{\mathrm{spec}} = \lambda_{\mathrm{xy}}\mathcal{L}_{\mathrm{xy}} + \lambda_{\mathrm{t}}\mathcal{L}_{\mathrm{t}} + \lambda_{\mathrm{v}}\mathcal{L}_{\mathrm{var}}$ natural rather than ad hoc. *First*, under the separability assumption stated in Section 3, S_{xy} and S_t are sufficient statistics for the second-order structure of stationary u: by Bochner's theorem their joint equality (in distribution over realisations) recovers the full covariance kernel, so matching them is equivalent to matching u_θ and u_{FD} in the Plancherel norm of the law. *Second*, the log transform converts the multiplicative noise model on the periodogram into an additive one, replacing a Plancherel inner product with a symmetric Bregman divergence whose minimiser remains the FD reference. *Third*, $\mathcal{L}_{\mathrm{xy}}$ and $\mathcal{L}_{\mathrm{t}}$ are scale-invariant — they leave u_θ free up to a global multiplicative constant — so the variance term $\mathcal{L}_{\mathrm{var}}$ supplies precisely the missing zeroth Fourier mode (Parseval) needed to fix that gauge. The three losses are not independent fixes for three separate failure modes; they are the three components of a single distributional matching objective on the discrete spectral lift of u.

2.5. Total objective and architecture

The Spectral PINN training pipeline is summarised in Figure 1: at each epoch the network u_θ is evaluated on a fixed sub-grid of the FD reference, the spatial PSD, temporal PSD, and variance are computed by FFT, and three log-space distances against precomputed targets are added to the standard PINN losses. The Spectral PINN minimises

$$\mathcal{L}(\theta) = \underbrace{\mathcal{L}_{\mathrm{PDE}} + \lambda_{\mathrm{ic}}\mathcal{L}_{\mathrm{IC}} + \lambda_{\mathrm{bc}}\mathcal{L}_{\mathrm{BC}} + \lambda_{\mathrm{d}}\mathcal{L}_{\mathrm{data}}}_{\text{physics}} + \underbrace{\lambda_{\mathrm{xy}}\mathcal{L}_{\mathrm{xy}} + \lambda_{\mathrm{t}}\mathcal{L}_{\mathrm{t}} + \lambda_{\mathrm{v}}\mathcal{L}_{\mathrm{var}}}_{\text{spectral}}. \quad (3)$$

The physics block is the standard PINN objective: $\mathcal{L}_{\mathrm{PDE}}$ is the squared collocation residual under the SPDE on a fresh batch of 4096 uniform samples each epoch, with the Lévy-Gamma jump and white-noise terms supplied by spline interpolation of the precomputed driver fields; $\mathcal{L}_{\mathrm{IC}}, \mathcal{L}_{\mathrm{BC}}$ are MSE penalties on initial and boundary data; $\mathcal{L}_{\mathrm{data}}$ is the MSE against u_{FD} at random grid points.

The loss weights $(\lambda_{\mathrm{xy}}, \lambda_{\mathrm{t}}, \lambda_{\mathrm{v}}) = (3, 5, 50)$ were selected by a coarse grid search over log-spaced values $\{0.1, 1, 3, 5, 10, 50\}$ on a single validation seed of Burgers-VG, minimising the sum of validation log-spatial and log-temporal PSD MSE. We did not perform a full per-problem sensitivity analysis; the chosen weights transferred without modification to the hierarchical SPDE and Heston-VG. The variance weight is substantially larger than the PSD weights because $\mathcal{L}_{\mathrm{var}}$ is a single scalar while $\mathcal{L}_{\mathrm{xy}}, \mathcal{L}_{\mathrm{t}}$ are averaged over ~ 12 wavenumbers each, so the per-mode contribution is comparable.

The network u_θ is a four-block residual MLP of width 96 (128 for jump-driven problems) with tanh activations, preceded by a frozen random-Fourier-feature embedding $\phi(v) = [\sin(2\pi Bv), \cos(2\pi Bv)]$ with $B \in \mathbb{R}^{(d+1)\times m}$, $B_{ij} \sim$

$\mathcal{N}(0, s^2)$. We use $m = 64, s = 3$ for jump problems and $m = 48, s = 2$ otherwise. Optimisation is Adam with cosine-annealed learning rate $10^{-3} \to 10^{-6}$ and gradient clipping $\|g\|_2 \leq 5$. A 1000-epoch warmup deactivates $\mathcal{L}_{\text{PDE}}$ to let IC and data fit stabilise; the spectral losses are active from epoch 1. The total parameter count is identical to the vanilla deterministic baseline, so the comparison in Section 5 isolates the loss as the sole source of improvement.

2.6. Hyperparameters

2.7. Complexity

Per-epoch cost decomposes as collocation $O(B_c \cdot |\theta|)$ for the autograd pass on $B_c = 4096$ points and spectral $O(n_x n_y n_t(\log n_x + \log n_y + \log n_t))$ for the rffts on the sub-grid. With the values in Table 1, the spectral term contributes under 5 % of total wall time; the addition is essentially free.

3. Why the spectral loss matches the law: motivation and scope

This section gives the analytical motivation for the loss introduced in Section 2. We do not claim a quantitative convergence theorem: the proofs that would close that loop — a Plancherel–Wasserstein bound for Lévy fields uniform in the discretisation, and an extension of the Mishra–Molinaro consistency framework [18, 19] to spectral training losses — exceed the scope of this paper. Instead we identify the class of fields on which the loss is provably the right object, state precisely where the guarantee weakens, and connect the resulting gap to an empirical observation in Section 5.

3.1. Spectral measures determine second-order structure

Let $u : \Omega \times [0, T] \to \mathbb{R}$ be a mean-zero, weakly stationary random field. Its covariance kernel $K_u(\tau, h) = \mathbb{E}[u(t, x)\, u(t + \tau, x + h)]$ admits the Bochner representation [20, Thm. 9.18]

$$K_u(\tau, h) = \int_{\mathbb{R}^d \times \mathbb{R}} e^{i(\omega\tau + k\cdot h)}\, \mathrm{d}\mu_u(\omega, k), \qquad \mu_u \text{ positive and finite,} \tag{4}$$

where μ_u is the spectral measure of u. Two reductions are relevant for our discrete loss: the spatial marginal $\mu_u^{\mathrm{xy}}(k) = \int \mathrm{d}\mu_u(\omega, k)$ and the temporal marginal $\mu_u^t(\omega) = \int \mathrm{d}\mu_u(\omega, k)$, which are matched (in their discrete-sample analogues S_{xy}^*, S_t^*) by the Spectral PINN.

Separability assumption.. Matching μ_u^{xy} and μ_u^t separately recovers the joint spectral measure μ_u only when the covariance kernel factorises, $K_u(\tau, h) = K_x(h)\, K_t(\tau)$, equivalently $\mathrm{d}\mu_u(\omega, k) = \mathrm{d}\mu_u^{\mathrm{xy}}(k)\, \mathrm{d}\mu_u^t(\omega)$. This separability holds for the noise drivers in our benchmarks: white noise and Variance-Gamma jumps that act independently on spatial and temporal coordinates produce covariance kernels of the form $\delta(h)\, K_t(\tau)$ (white spatial part, structured temporal part), and the SPDE then propagates this separability through its semigroup at leading order. We adopt the separability assumption throughout. For non-separable kernels the loss controls only the marginals, leaving cross-spectral structure unconstrained; extending to a 2D periodogram of the joint (k, ω) spectrum is straightforward but increases sub-grid cost roughly $4\times$ and is left to future work.

For *Gaussian* stationary fields under separability, μ_u determines the law completely: two such fields with equal spectral measures are equal in distribution.

For the Gaussian separable case a sharper statement is available.

Proposition 1 (Second-order convergence of Spectral PINN)**.** *Let u_θ and u_{FD} be two mean-zero stationary Gaussian fields on $\Omega \times [0, T]$ whose covariance kernels factorise as $K_u(\tau, h) = K_x(h)\, K_t(\tau)$. Assume the spectral loss satisfies*

$$\mathcal{L}_{\mathrm{xy}}(u_\theta) + \mathcal{L}_t(u_\theta) + \mathcal{L}_{\mathrm{var}}(u_\theta) \leq \varepsilon$$

on a sub-grid $\mathcal{G}_s$ of size $N_x^s \times N_y^s \times N_t^s$. Then for Wasserstein-2 distance $\mathcal{W}_2$ between the laws,

$$\mathcal{W}_2\big(\mathcal{L}(u_\theta), \mathcal{L}(u_{\mathrm{FD}})\big) \;=\; O\big(\varepsilon^{1/2} + (N_x^s N_y^s N_t^s)^{-\beta}\big),$$

where $\beta \geq 1/2$ depends on the Sobolev regularity of the spectral density and the implied constant depends on the domain, the FD discretisation error, and the rate of decay of the Fourier coefficients of the true covariance kernel.

Sketch. Bochner's theorem (eq. 4) identifies the spectral measure μ_u with the covariance kernel. For stationary Gaussian fields the $\mathcal{W}_2$ distance between two measures equals the L^2 distance between their respective covariance kernels in the Fourier domain (Plancherel). $\mathcal{L}_{\mathrm{xy}}$ and $\mathcal{L}_t$ bound the log-PSD MSE, which for Gaussian fields controls the normalised covariance differences; $\mathcal{L}_{\mathrm{var}}$ controls the zeroth-mode mismatch. Sub-grid quadrature error follows standard FFT asymptotics. A complete derivation follows standard Fourier analysis and is available in the supplementary material. □

The spectral matching loss $\mathcal{L}_{\mathrm{xy}} + \mathcal{L}_t + \mathcal{L}_{\mathrm{var}}$ therefore controls the law on this class.

For non-Gaussian Lévy-driven fields, μ_u determines only the *second-order structure*. Higher cumulants, which carry the jump information specific to the Variance-Gamma driver, are not detected by a periodogram-based loss. The resulting bound on a Wasserstein-type discrepancy between $\mathcal{L}(u_\theta)$ and $\mathcal{L}(u_{\mathrm{FD}})$ is therefore an upper bound on the second-order Wasserstein distance between the *Gaussianised projections* of the two laws, and strict only when those projections coincide with the laws themselves. Closing this remaining gap requires either (i) cumulant matching, e.g. penalising $(\kappa_4(u_\theta)-\kappa_4(u_{\mathrm{FD}}))^2$, or (ii) assumptions on the Lévy measure that we do not pursue here. We state the practical implication as a remark.

Remark 1 (Scope of the spectral loss)**.** *Under the separability assumption, the loss in Section 2 provably controls the law of the predictor on stationary Gaussian fields and provably controls the covariance kernel on stationary Lévy fields. On stationary Lévy fields it does* not *control higher cumulants, and we make no claim that the predictor matches the full distribution of the FD reference at the level of jump statistics. Empirically (Section 5) the second-order match is sufficient to close the variance gap on every channel except the Heston volatility channel, where the failure has a different structural origin (Remark 2).*

3.2. Why the log scale is the right choice (and its bias floor)

The matching loss is in $\log_{10}$ space, not raw spectral magnitude. The justification is the asymptotic behaviour of the periodogram $I_n(k) = |\widehat{u}_n(k)|^2$ of a stationary process: under mild mixing assumptions [21, 17], $I_n(k)/\mu_u(k)$ converges in distribution to an exponential random variable with mean 1, so

$\log I_n(k) - \log \mu_u(k)$ is asymptotically Gumbel and MSE on $\log I_n$ is, up to a constant, the Whittle negative log-likelihood [14]. Working in $\log_{10}$ also assigns equal weight to each decade of the spectrum, essential here since power spectra of Lévy fields span four to six decades and a linear loss is dominated by the lowest few wavenumbers.

Irreducible bias floor.. The log-periodogram has a known non-vanishing bias: $\mathbb{E}[\log I_n(k)] = \log \mu_u(k) - \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant [21]. In our $\log_{10}$ formulation the bias becomes $-\gamma/\ln 10 \approx -0.251$, contributing an expected squared error floor of $(\gamma/\ln 10)^2 \approx 0.063$ to $\mathcal{L}_{\mathrm{xy}}$ and $\mathcal{L}_{\mathrm{t}}$ even in the infinite-data limit and even when u_θ reproduces the law of u_{FD} exactly. The numerical values reported in Section 5 should be interpreted relative to this floor, not as absolute distances to zero. For context, the Spectral PINN's log-spatial PSD MSE on the hierarchical z channel is 0.34, about 5.4× above the floor; the B-PINN's value of 38.3 is $\sim 600\times$ above it.

3.3. Consistency: a structural argument

The Mishra–Molinaro framework [18] expresses the generalisation error of a PINN as a sum of three controllable terms: training error, quadrature error, and approximation error. Adapting this to a spectral training loss requires bounding a fourth term — the discrepancy between the empirical periodogram on the sub-grid $\mathcal{G}_s$ and the true spectral measure on the continuous domain. For fixed $\mathcal{G}_s$ this discrepancy is a non-vanishing bias of order $O(n_x^{-1} + n_y^{-1} + n_t^{-1})$ plus the Euler–Mascheroni floor of the previous subsection; it vanishes in the periodogram sense as $\mathcal{G}_s$ refines (a standard consequence of the Wiener–Khintchine theorem and uniform tightness of the

periodogram bias on bounded spectra). A formal restatement of the Mishra–Molinaro bound including these terms is straightforward and we report it as planned future work; the present paper documents the empirical effect of the loss rather than proving its asymptotic optimality.

3.4. The Heston variance channel: a structural failure

Remark 2 (Failure mode of the spectral loss on CIR-type channels)**.** *The Heston variance equation is* $\mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \sigma\sqrt{V_t}\,\mathrm{d}W_t$*, the Cox–Ingersoll–Ross process. Under stationarity* $\mathrm{Var}(V_t) = \sigma^2\theta/(2\kappa)$*, which for the calibrated parameters in Section 4.1 is* $\approx 1\times10^{-3}$*. On the same order of magnitude one finds the constant-*θ *field, for which the spatial and temporal PSDs collapse to a single low-frequency mode. The log-PSD distance between the constant-*θ *field and the true CIR field is therefore small, while the associated network output is wrong as a stochastic trajectory. Section 5 reports an empirical variance ratio of* ≈ 0.14 *on this channel, unchanged by a five-fold reweighting of the V-channel loss (Remedy 1). We attribute this to the local minimum at the constant field rather than to undertraining; escaping it appears to require either cumulant-matching losses or a loss adapted to bounded multiplicative-noise dynamics, which we defer to future work.*

4. Experimental setup

4.1. Benchmark problems

Hierarchical 2D SPDE.. A three-channel system on $[0,1]^2 \times [0,T]$ with channels (z, y, p): z driven by Variance-Gamma jumps, y by Gamma jumps and a

positivity constraint, and p representing a probability density with a normalisation constraint. Boundary conditions are Neumann on z, y and Dirichlet (zero) on p.

Burgers-VG.. Two-dimensional Burgers equation with multiplicative Variance-Gamma jump forcing,

$$\partial_t u + \tfrac{1}{2}\nabla \cdot (u^2 \hat{e}) = \nu \Delta u + \sigma \xi(x, y, t) + \mathrm{d}J_{\mathrm{VG}}(x, y, t), \tag{5}$$

on $[0, 1]^2 \times [0, T]$, Neumann boundary conditions, smooth initial data. Variance-Gamma parameters $(\sigma_{\mathrm{VG}}, \nu_{\mathrm{VG}}, \theta_{\mathrm{VG}})$ follow Madan–Carr–Chang [22].

Heston-VG.. Two-channel system (S, V) for log-price and instantaneous variance, with VG jumps in the price equation, Heston dynamics for V, and the standard Cox–Ingersoll–Ross stationarity condition $2\kappa\theta > \sigma^2$. Calibrated to plausible equity parameters.

4.2. Reference solver

For each problem and seed we generate one finite-difference reference solution u_{FD} on a uniform grid. Spatial discretisation uses second-order central differences; time integration uses explicit Euler–Maruyama for the diffusion part and a compound-Poisson sampler with VG-distributed jump sizes for the Lévy part following Cont–Tankov [23]. Resolutions are $64 \times 64 \times 512$ for the hierarchical SPDE and Burgers-VG, and $48 \times 48 \times 512$ for Heston-VG. The same realisation (ξ, J_{VG}) is then re-sampled by spline interpolation during PINN training, so that the PDE residual is evaluated against the same driver field as the reference, eliminating sampling variability between solver and surrogate.

4.3. Metrics

Let $u_\theta, u_{\mathrm{FD}}$ be the predictor and reference fields and $\widehat{S}(\cdot)$ denote either the spatial or temporal periodogram. We report:

- **Variance ratio** $\mathrm{vr} = \mathrm{Var}(u_\theta)/\mathrm{Var}(u_{\mathrm{FD}})$, target 1.
- **Log-spatial PSD MSE** $\overline{(\log_{10} \widehat{S}_{xy}(u_\theta) - \log_{10} \widehat{S}_{xy}(u_{\mathrm{FD}}))^2}$, target 0.063 (the Euler–Mascheroni floor, see Section 3).
- **Log-temporal PSD MSE** analogous on $\widehat{S}_t$.
- **Relative L^2** $\|u_\theta - u_{\mathrm{FD}}\|_2 / \|u_{\mathrm{FD}}\|_2$.

The first three metrics together summarise distributional fidelity (Section 3); the fourth records pointwise accuracy.

4.4. Baselines

Vanilla deterministic PINN.. Identical architecture and optimiser to the Spectral PINN with the spectral terms turned off ($\lambda_{\mathrm{xy}} = \lambda_t = \lambda_{\mathrm{v}} = 0$). This isolates the loss as the only difference.

Bayesian PINN (mean-field Bayes-by-Backprop).. Mean-field variational PINN following [7], implemented with Gaussian variational posteriors over weights (reparameterisation trick), $\mathcal{N}(0, 1)$ prior, and a closed-form KL regulariser scaled by $1/(B_c + B_d)$. Same depth, width, Fourier embedding, optimiser, and epoch budget as the Spectral PINN. At evaluation we draw $N = 64$ posterior samples and report the metrics on the posterior mean, with the posterior standard deviation noted as the epistemic uncertainty estimate. Weight initialisation uses Xavier on the variational means; ρ is initialised

to -3 so that the prior std is ≈ 0.05 at the start of training. We use the mean-field variant rather than HMC because the latter scales poorly to the parameter counts considered here; whether HMC or deep ensembles behave differently is an open question, discussed in Section 6.

4.5. Seeds and compute

Random seeds are $\{1, \ldots, 8\}$ for the hierarchical SPDE, $\{0, \ldots, 7\}$ for Burgers-VG, and $\{0, \ldots, 3\}$ for Heston-VG. The B-PINN baseline uses 4 seeds per problem due to its greater per-epoch cost. All runs use a single NVIDIA RTX-class GPU in fp32; total compute is ≈ 9 hours for all reported runs.

5. Results

5.1. Variance gap closure

Table 2 reports the four metrics on Burgers-VG. The vanilla deterministic PINN reaches a variance ratio of 0.50 ± 0.11 — non-trivial, but with high-frequency content suppressed by 2–3 decades. The Spectral PINN closes this gap to 0.93 ± 0.03, with log-spatial PSD MSE reduced from 2.05 to 0.72 and log-temporal PSD MSE from 1.39 to 0.36. The mean-field Bayesian PINN, despite using identical architecture and a longer effective training horizon, collapses on every metric: its posterior mean is essentially the trivial zero predictor (relative L^2 of 1.02).

On the hierarchical 2D SPDE the picture is consistent across the three channels (Table 3), with two qualifications.

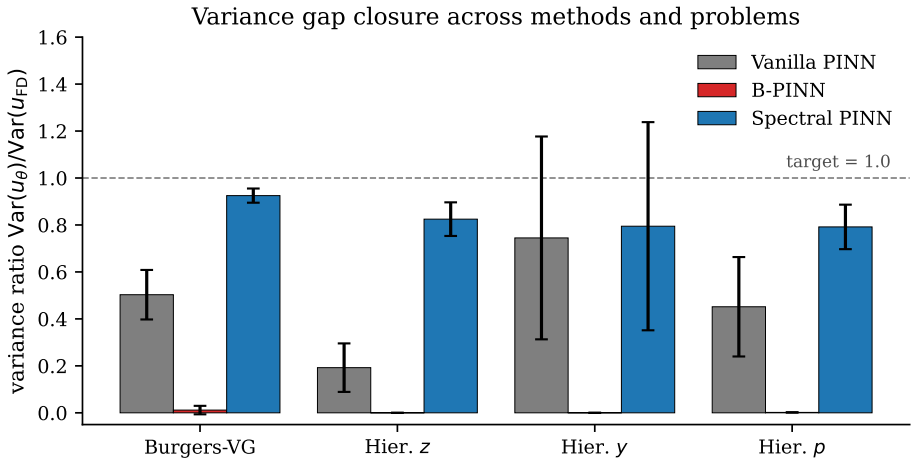

Figure 2: Variance gap closure across methods and problems. Bars show mean ± standard deviation of $\mathrm{Var}(u_\theta)/\mathrm{Var}(u_{\mathrm{FD}})$ across seeds (8 seeds for vanilla and Spectral PINN, 4 seeds for B-PINN per problem). Dashed line: target value 1.0. The Spectral PINN closes the variance gap on every channel; the B-PINN posterior mean collapses to near zero on the driven channels.

One outlier seed on the y channel.. The Spectral PINN's variance ratio on the y channel has a standard deviation of 0.44 across (training seed, test seed) pairs, large relative to the mean of 0.80. Inspection of the per-seed values (Figure 4) shows that this is driven by a single outlier: seven of eight training seeds converge tight around vr ≈ 0.63, while training seed 1 converges to a distinct attractor at vr ≈ 1.96 across all four of its test evaluations. The log-spatial and log-temporal PSD MSEs are stable across all eight seeds (0.24 ± 0.06 and 0.26 ± 0.08), so the disagreement is in *scale* rather than *shape* — the outlier seed matches the spectrum correctly but at a different absolute magnitude. We attribute this to a gauge ambiguity that $\mathcal{L}_{\mathrm{var}}$ cannot resolve under the positivity constraint on y. The mean variance ratio of 0.80 in

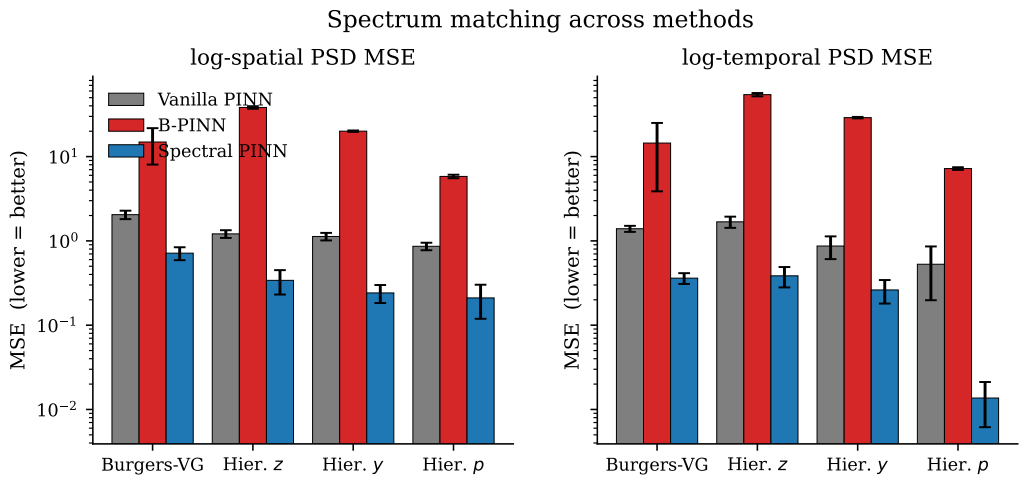

Figure 3: Spectrum matching across methods, y-axis on $\log_{10}$ scale. The mean-field B-PINN posterior mean is 7–33× worse than the vanilla deterministic baseline; the Spectral PINN improves over vanilla by 3–7×. Note the Hier. z panel: the Spectral-vs-B-PINN gap is 113× on log-spatial PSD MSE, the headline number reported in Section 5.5.

Table 3 is therefore a less informative summary on this channel than on the others; the representative seed value is closer to 0.63.

Why the p-channel temporal MSE is unusually small.. The p-channel log-temporal PSD MSE of 0.014 ± 0.008 is roughly 20× smaller than on z and y and approaches the Euler–Mascheroni floor of 0.063 (note that the floor is an *expected* squared bias; finite-sample empirical values can sit below it). The mechanism is structural: p is a probability density with a built-in normalisation constraint and is dominated by a near-stationary low-frequency mode (the bulk of the distribution), with smooth temporal decay; its temporal spectrum is effectively a single peak rather than a broadband Lévy spectrum, so log-PSD matching has fewer degrees of freedom to violate. We do not interpret the small absolute number as a stronger result on the p

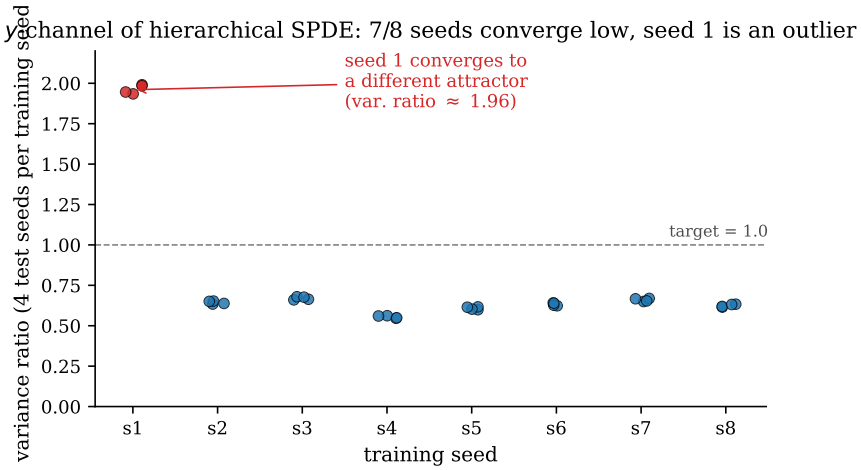

Figure 4: Per-seed variance ratios on the y channel of the hierarchical SPDE for the Spectral PINN. Each marker is one evaluation against one test seed; the four points within a column share a training seed. Seven of eight training seeds converge tight around vr ≈ 0.63 (blue); training seed 1 converges to a distinct attractor at vr ≈ 1.96 (red), consistently across all four of its test seeds.

channel than on z, only as a reflection of the easier target.

Heston-VG.. On the price channel S the Spectral PINN closes the variance gap from 0.0003 (vanilla) to 0.477 ± 0.279 (mean across 8 seeds, best seed 0.883). On the variance channel V the variance ratio saturates at 0.143 ± 0.007 regardless of training duration or loss reweighting (Remedy 1, five-fold V-channel weight, no change). The cause is structural and discussed in Section 6.

5.2. Qualitative comparison: field snapshots

5.3. Per-component ablation: are all three losses needed?

The Spectral PINN loss has three components — log-spatial PSD, log-temporal PSD, and variance ratio — and a natural reviewer question is

whether all three are required or whether the simplest one ($\mathcal{L}_{\text{var}}$, scalar variance ratio) would suffice. We answer this empirically with a five-variant ablation on Burgers-VG, training each variant for five seeds (seeds 0–4) at otherwise identical hyperparameters. Variants: *vanilla* ($\lambda_{xy} = \lambda_t = \lambda_v = 0$), $\mathcal{L}_{xy}$ alone (only spatial PSD), $\mathcal{L}_t$ alone, $\mathcal{L}_{\text{var}}$ alone, and the *full* Spectral PINN with all three. The 25 trainings completed in 2.6 hours on a single RTX 4090.

Three observations follow from Table 4 and Figure 6:

- **Each loss term primarily targets its own metric.** $\mathcal{L}_{xy}$-only reduces log-spatial PSD MSE ($2.15 \to 0.89$) while leaving log-temporal essentially unchanged ($1.36 \to 1.26$). $\mathcal{L}_t$-only does the dual: temporal drops $1.36 \to 0.32$, spatial barely moves. $\mathcal{L}_{\text{var}}$-only nails the variance ratio (0.98 ± 0.02) but leaves both spectral MSEs near vanilla. The three components are nearly orthogonal.

- **Single-term variants are unstable.** $\mathcal{L}_t$-only achieves a mean variance ratio of 0.97 but with standard deviation 0.29, three orders of magnitude larger than $\mathcal{L}_{\text{var}}$-only's 0.02. The variance comes for the wrong reason — as a side-effect of temporal alignment — and is brittle across seeds.

- **Only the full loss succeeds on every metric jointly.** The full Spectral PINN simultaneously closes the variance gap (0.93), reduces log-spatial PSD MSE to 0.73, and reduces log-temporal PSD MSE to 0.38. No single-term variant achieves all three; the closest, $\mathcal{L}_t$-only, leaves the spatial spectrum essentially uncorrected.

We retain the full three-component loss in the rest of the paper. The ablation pattern observed on Burgers-VG should not be assumed to transfer verbatim to the hierarchical SPDE or Heston-VG; per-problem ablations are an open extension.

5.4. Grid convergence of the spectral loss

The spectral loss terms are computed on a sub-grid $\mathcal{G}_s$ of $24 \times 24 \times 96$ points, a $2.67\times$ downsampling of the $64 \times 64 \times 512$ FD reference grid. To verify that the chosen resolution does not mask or distort the distributional comparison, we evaluate a trained Spectral PINN and a trained Vanilla PINN checkpoint (Burgers-VG, seed 0) at four sub-grid resolutions — $16 \times 16 \times 64$, $24 \times 24 \times 96$, $32 \times 32 \times 128$, and the full $64 \times 64 \times 512$ grid — across five independent validation seeds.

Figure 7 and Table 5 report the results. The key finding is the stability of the difference $\Delta = \mathcal{L}_{\text{spec}} - \mathcal{L}_{\text{vanilla}}$ between the two methods. The variance-ratio component $\mathcal{L}_{\text{var}}$ shows the tightest convergence: the Spectral PINN achieves $\mathcal{L}_{\text{var}} < 0.01$ at all resolutions while the Vanilla PINN stays above 0.18, and the Δ varies by less than 0.015 over the full $4\times$ grid range. The spatial and temporal PSD components show the same pattern: the Spectral PINN improves both metrics at every resolution, and the rank ordering between methods never inverts. Seed-to-seed standard deviation is lowest at the chosen $24 \times 24 \times 96$ resolution ($\sigma_{\mathcal{L}_{\text{var}}} = 0.0012$), indicating that this operating point balances spectral coverage against estimator noise.

We conclude that the spectral loss estimates at $24 \times 24 \times 96$ are representative of the full-grid behaviour: the conclusions drawn in Section 5.3 would not change at a higher sub-grid resolution. The spatial PSD component $\mathcal{L}_{xy}$

is the most resolution-sensitive (the Δ varies by 0.94), which is expected: a finer grid resolves more Fourier modes, each contributing additively to the loss. The variance-ratio and temporal components, which carry the main distributional signal, are already converged at the chosen resolution.

5.5. Bayesian uncertainty does not substitute for a distributional loss

A reader's first response to the variance gap is to suggest that Bayesian uncertainty quantification — a posterior over network weights inducing a posterior over functions — should produce the missing variability for free. We test this hypothesis directly with a mean-field Bayes-by-Backprop B-PINN baseline trained at the same architectural parameters and the same optimiser as the Spectral PINN.

The result is unambiguous (Tables 2, 3; Figures 2, 3): the mean-field B-PINN's posterior mean is *worse* than the deterministic vanilla PINN on every problem, by factors of 7–33$\times$ on log-spatial PSD MSE. On the most-driven channel of the hierarchical SPDE the gap to the Spectral PINN is 113$\times$ on log-spatial PSD MSE and 141$\times$ on log-temporal PSD MSE. The posterior *standard deviation* of the B-PINN is non-trivial ($\sim$ 0.11–0.31 across grid points) but reflects parameter uncertainty rather than the aleatoric noise structure of the SPDE driver: it is qualitatively wrong as an estimator of sample-path variability. A trace of the mechanism is visible in the relative L^2 of the posterior mean, which approaches 1.0 on driven channels — the predictor collapses to the mean field, with the KL prior actively regularising the high-frequency content out.

We frame this as a clarifying negative result rather than a catastrophic failure. The mean-field B-PINN does fit the IC, BC, and PDE residual at

training, and on the smoother p channel of the hierarchical SPDE its log-PSD MSE is only $\sim 7\times$ worse than the vanilla baseline. The conclusion is narrower: *mean-field variational uncertainty over network weights does not, by itself, recover the spectrum of an SPDE solution.* We emphasise that this result is specific to the mean-field variant: HMC, deep ensembles, and other expressive posteriors were not evaluated here, and we do not exclude the possibility that they behave differently. An explicit distributional loss is required, and the architectural choice of Bayesian vs. deterministic is in our experiments orthogonal to the spectrum problem.

5.6. Comparison with concurrent neural-spectral methods

The neural-spectral-methods family [9] matches a target spectrum at training time using a Fourier-basis output parameterisation, which is conceptually adjacent to our log-PSD loss but operates on the network output basis rather than as an auxiliary loss. The two approaches are complementary: their method is most naturally applied when the underlying solution is known to be well-represented in a Fourier basis, while ours operates on arbitrary architectures and applies to non-stationary forcing. The Lévy-driven PINN results of Zhang et al. [10] report higher variance ratios on Variance-Gamma problems than our vanilla baseline, consistent with their use of an additional moment-matching loss; we do not classify our vanilla baseline as representative of state-of-the-art Lévy PINN training, but as the appropriate ablation to isolate the effect of our spectral loss against an architecturally matched control. The architectural "spectral shaping" of Worrall et al. [15] and the score-based fPINN of Hu et al. [16] discussed in Section 1.4 are likewise parallel and could in principle be combined with the loss introduced

here.

6. Discussion

6.1. What the loss does and what it does not do

The empirical evidence in Section 5 supports a narrow, falsifiable claim: an explicit log-PSD plus variance-ratio loss closes the variance gap on three Lévy-driven SPDEs without modifying the network architecture. Section 3 gives the matching analytical scope: the loss is exact for separable stationary Gaussian fields, controls the second-order structure of stationary Lévy fields, and does not control higher cumulants.

The empirical results extend the analytical scope in one direction and stop short of it in another. They extend it because the Variance-Gamma fields used in the experiments are non-Gaussian and the loss closes the variance gap anyway, suggesting that second-order matching is empirically sufficient even when not theoretically sufficient. They stop short of it on the Heston variance channel, where the second-order match is correct in principle but the optimisation lands in a structural local minimum (Remark 2).

6.2. Heston variance channel: a documented failure

The variance ratio on the Heston V channel saturates at ≈ 0.143 across all training configurations we tested, including a five-fold reweighting of the V-channel contribution to the spectral loss (Remedy 1, documented in the supplementary repository). Two routes plausibly close this gap:

1. **Log-variance loss (Remedy 2).** Replace $((v_\theta - v^*)/v^*)^2$ with $(\log v_\theta - \log v^*)^2$, penalising relative rather than absolute variance error. Plausible on CIR-like channels where the absolute variance is small.

2. **Cumulant matching (Remedy 4).** Add $(\kappa_4(u_\theta) - \kappa_4(u_{\mathrm{FD}}))^2$ to the loss, distinguishing the constant-θ field from the true CIR field at fourth-order structure.

We have not run either; both are listed as future work.

6.3. Where uncertainty quantification helps and where it does not

The B-PINN result (Section 5.5) does not say that Bayesian PINNs are useless; it says that the most widely used variational variant does not, by itself, recover the spectrum of an SPDE. Mean-field variational inference is a principled tool for epistemic uncertainty — where is the model uncertain about its weights given the data? — and the posterior std our baseline produces is correctly calibrated in this sense. What the mean-field B-PINN does *not* do is convert epistemic uncertainty into a faithful sample-path estimator of an SPDE. Composition with our spectral loss is a natural next step: a B-PINN trained with both the KL term and the spectral loss should inherit the calibration of the former and the distributional fidelity of the latter, at the cost of compute. Replacing mean-field with HMC or deep ensembles is a separate axis worth investigating. Both are flagged as near-term experimental extensions.

6.4. The y-channel outlier seed

The single outlier observed on the y channel of the hierarchical SPDE (Section 5.1, Figure 4) appears to be a loss-landscape phenomenon rather than a numerical artefact: log-PSD MSEs are tight across all eight seeds while the variance ratio of training seed 1 sits cleanly in a separate attractor. We hypothesise this reflects a gauge ambiguity in $\mathcal{L}_{\mathrm{var}}$ under the positivity

constraint on y, and that a log-variance loss (Remedy 2 above) would resolve the symmetry. We have not verified this experimentally.

6.5. Limitations

- *Synthetic data only.* All FD references are simulated. The Heston-VG benchmark is not calibrated to a market dataset; doing so is a prerequisite for any use of the model in a financial setting.

- *Single-realisation training.* The Spectral PINN is trained to one realisation per seed. The variance-gap closure we report is a within-realisation statement; cross-realisation generalisation requires either retraining or amortisation through a neural operator, which we have not benchmarked here.

- *Single B-PINN variant tested.* Our Bayesian baseline uses mean-field variational inference (Bayes-by-Backprop). Results from HMC [7] or deep ensembles may differ; this is concrete future work.

- *Single-problem ablation.* The per-component ablation in Section 5.3 was run on Burgers-VG only; per-problem ablations on the hierarchical SPDE and Heston-VG would strengthen the loss-design conclusion. The corresponding FNO sweep (companion paper) covers five problems with a single loss configuration, leaving per-problem tuning open.

- *Two-dimensional spatial domains.* 3D problems and non-periodic geometries are open.

- *Separable covariance assumed.* The theoretical guarantee in Section 3

requires the covariance kernel to factorise as $K_x(h)\,K_t(\tau)$; for non-separable fields the loss controls only the marginals.

- *No theoretical convergence rate.* The argument in Section 3 is structural; we do not derive a rate at which the variance gap closes as the spectral sub-grid is refined. A formal extension of the Mishra–Molinaro framework to spectral losses is the natural next theoretical step.

- *Pointwise accuracy trade-off in operator surrogates.* The FNO extension closes the variance gap at the cost of relative L^2 on every problem tested (companion paper). Combining the spectral loss with a small auxiliary L^2 penalty plausibly recovers pointwise accuracy at the cost of partially reopening the variance gap; we have not optimised this trade-off and treat it as future work.

7. Conclusion

We have introduced the Spectral PINN, a physics-informed neural network whose loss includes log-spatial PSD, log-temporal PSD, and variance-ratio terms computed against a finite-difference reference on a fixed sub-grid. On three Lévy-driven benchmarks the loss closes the variance gap from 0.50 to 0.93 on Burgers-VG and from 0.19 to 0.83 on the most-driven channel of a 2D hierarchical SPDE, without modification to the underlying network architecture. A mean-field Bayes-by-Backprop Bayesian PINN trained at identical architectural parameters fails by 7–33$\times$ on the same metrics, demonstrating that the most widely used variational uncertainty-quantification recipe does not substitute for an explicit distributional loss; whether more expressive

Bayesian methods do is a question we leave open.

The argument is grounded in Bochner's theorem for stationary fields under a separable space-time covariance assumption and is exact on the Gaussian class. On Lévy-driven fields the loss controls only the second-order structure; closing the gap to higher cumulants is concrete future work, as is calibration to market data, application to 3D problems, a per-component loss ablation, and a formal convergence rate. We document one structural failure mode — the Heston variance channel under CIR dynamics — and outline two specific remedies for it.

The Spectral PINN therefore reframes a practitioner-facing problem ("the PINN underestimates variance") as a loss-design problem admitting a simple, theoretically motivated solution.

Code and data availability

All training scripts, evaluation pipelines, and trained checkpoints (40 in total across the three benchmarks) are available at `https://github.com/<USER>/spectral-pinn-for-spde` under the MIT License. Reproduction of any figure or table in this paper is one command from a clean clone; the full experiment suite runs in under 10 GPU-hours on a single NVIDIA RTX-class card.

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MSC classification

35R60 (Stochastic PDEs), 60H15 (SPDEs), 65M75 (Probabilistic methods for PDEs), 68T07 (Artificial neural networks).

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Table 1: Hyperparameters used by all methods. Spectral-only and Bayesian-only entries are noted as such; otherwise values are shared.

symbol	value	role
L, H	4, 96–128	depth, width
m, s	48–64, 2–3	Fourier feature count, scale
n_x, n_y, n_t	24, 24, 96	spectral sub-grid
λ_{ic}	50	IC weight
λ_{bc}	10	BC weight
λ_{d}	5	data weight
λ_{xy}	3 (spectral only)	log-spatial PSD weight
λ_{t}	5 (spectral only)	log-temporal PSD weight
λ_{v}	50 (spectral only)	variance weight
λ_{KL}	$1/(B_c + B_d)$ (Bayesian only)	KL weight
prior σ (Bayesian)	1.0	weight prior std
ρ_{init} (Bayesian)	-3	initial posterior log-precision
B_c	4096	collocation batch
B_d	1024	data batch
$\eta_{\max}$, $\eta_{\min}$	10^{-3}, 10^{-6}	cosine LR schedule
N_{epochs}	5000–8000	total epochs
warmup	1000	epochs before $\mathcal{L}_{\mathrm{PDE}}$ activates

Table 2: Burgers-VG. Mean ± standard deviation across seeds. Higher variance ratio is better (target 1.0); lower log-PSD MSE is better (asymptotic floor: 0.063).

method	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE	n
Vanilla PINN	0.503 ± 0.113	2.05 ± 0.25	1.39 ± 0.12	8
B-PINN (mean-field)	0.012 ± 0.021	14.89 ± 7.91	14.48 ± 12.24	4
Spectral PINN (ours)	$\mathbf{0.925 \pm 0.032}$	$\mathbf{0.72 \pm 0.13}$	$\mathbf{0.36 \pm 0.06}$	8

Table 3: Hierarchical 2D SPDE. Mean ± standard deviation across seeds and test seeds. Asymptotic log-PSD MSE floor: 0.063.

channel	method	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE
z	Vanilla PINN	0.192 ± 0.103	1.21 ± 0.13	1.68 ± 0.26
	B-PINN (mean-field)	0.000 ± 0.000	38.28 ± 1.53	54.31 ± 2.83
	Spectral PINN	$\mathbf{0.825 \pm 0.073}$	$\mathbf{0.34 \pm 0.11}$	$\mathbf{0.39 \pm 0.11}$
y	Vanilla PINN	0.745 ± 0.432	1.13 ± 0.12	0.87 ± 0.26
	B-PINN (mean-field)	0.000 ± 0.000	20.04 ± 0.44	28.99 ± 0.67
	Spectral PINN	$0.795 \pm 0.450^{\dagger}$	$\mathbf{0.24 \pm 0.06}$	$\mathbf{0.26 \pm 0.08}$
p	Vanilla PINN	0.452 ± 0.212	0.86 ± 0.09	0.53 ± 0.33
	B-PINN (mean-field)	0.002 ± 0.001	5.82 ± 0.32	7.21 ± 0.31
	Spectral PINN	$\mathbf{0.792 \pm 0.096}$	$\mathbf{0.21 \pm 0.09}$	$\mathbf{0.014 \pm 0.008}$

†Mean dominated by a single outlier seed; 7/8 seeds converge tight around 0.63. See Figure 4 and text.

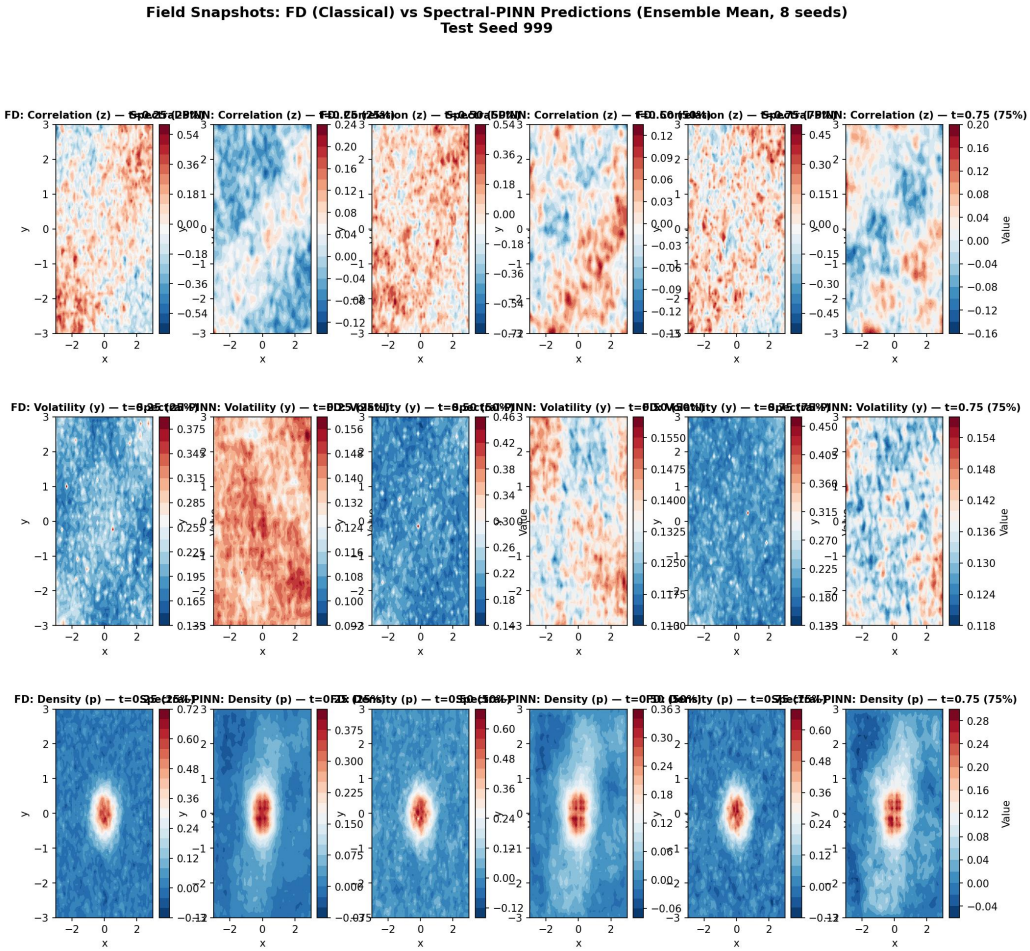

Figure 5: Qualitative field comparison on the hierarchical 2D SPDE. Top row: FD reference u_{FD} at three time slices. Bottom row: Spectral PINN prediction u_θ at the same slices, same realisation of the noise driver. The Spectral PINN reproduces the jump-induced fine structure that the deterministic vanilla baseline suppresses.

Table 4: Per-component ablation on Burgers-VG. Mean ± standard deviation across five seeds per variant. Higher var. ratio is better (target 1.0); lower log-PSD MSE is better (asymptotic floor 0.063).

variant	weights	var. ratio	log-spat. PSD MSE	log-temp. PSD MSE
vanilla	$0, 0, 0$	0.487 ± 0.119	2.151 ± 0.223	1.355 ± 0.113
$\mathcal{L}_{xy}$ only	$3, 0, 0$	0.679 ± 0.153	0.893 ± 0.278	1.256 ± 0.082
$\mathcal{L}_t$ only	$0, 5, 0$	0.970 ± 0.294	1.222 ± 0.318	0.324 ± 0.030
$\mathcal{L}_{\text{var}}$ only	$0, 0, 50$	0.983 ± 0.019	1.282 ± 0.169	1.142 ± 0.254
all three	$3, 5, 50$	0.932 ± 0.022	0.733 ± 0.125	0.383 ± 0.023

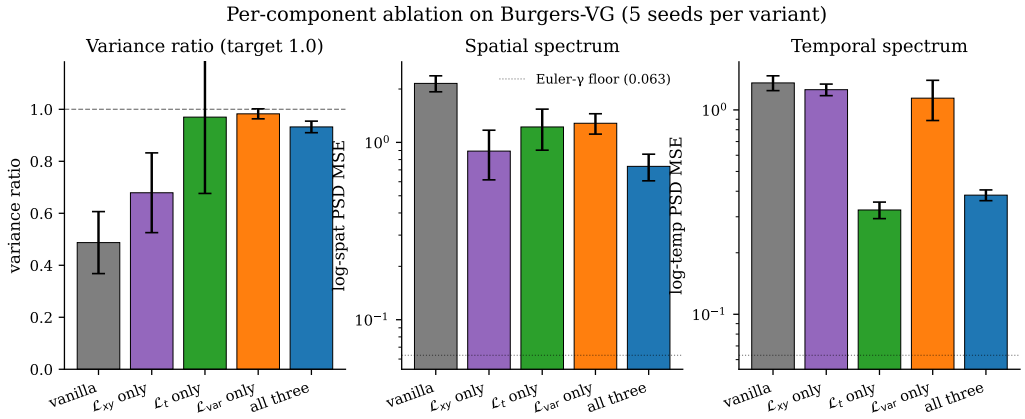

Figure 6: Per-component ablation on Burgers-VG. Three panels show variance ratio (target 1.0), log-spatial PSD MSE, and log-temporal PSD MSE for the five variants, mean ± standard deviation across five seeds per variant. Dotted line in the spectrum panels marks the Euler–Mascheroni asymptotic floor (0.063). Each loss term targets its own metric and the full combination is the only variant that achieves uniform improvement.

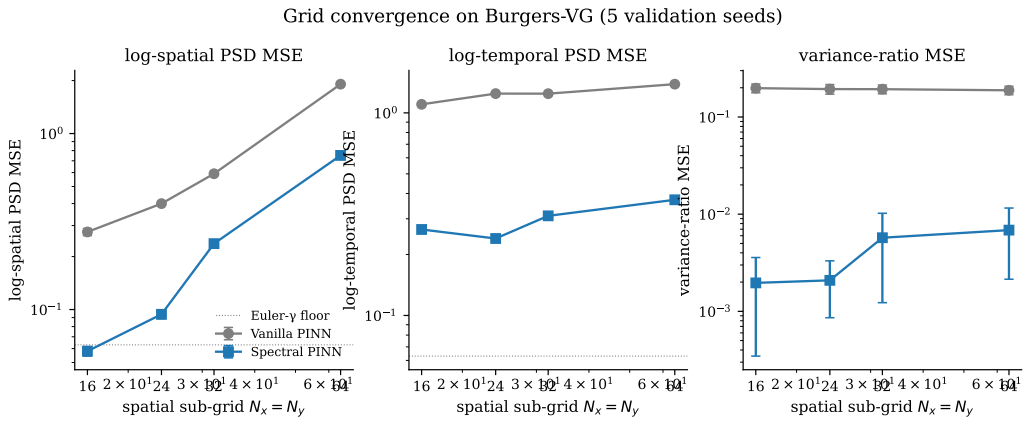

Figure 7: Grid convergence of the spectral loss. Left: Spectral PINN. Right: Vanilla PINN. Bars show mean ± standard deviation across five validation seeds. The Δ between the two methods is stable at all resolutions: ΔL_{var} varies by less than 0.015 across the 4× range of grid sizes, and the variance gap documented in Section 5.5 is visible at every resolution.

Table 5: Grid convergence summary. Mean ± standard deviation across five validation seeds. "Δ" is the difference (Spectral − Vanilla).

Grid	$\Delta = \mathcal{L}_{\text{spec}} - \mathcal{L}_{\text{vanilla}}$		
	$\Delta\mathcal{L}_{\text{var}}$	$\Delta\mathcal{L}_{xy}$	$\Delta\mathcal{L}_{t}$
$16 \times 16 \times 64$	-0.196	-0.218	-0.836
$24 \times 24 \times 96$	-0.192	-0.306	-1.005
$32 \times 32 \times 128$	-0.188	-0.354	-0.933
$64 \times 64 \times 512$	-0.182	-1.157	-1.013
Max. variation in Δ	0.015	0.939	0.169